



Data scientist take home technical test



You are given a dataset composed of images of bacteria such as the one attached (which is of course not a real image of bacteria but a drawing, for data confidentiality matters).

On this simplified image, we drew the following colors:

- blue and black, labelled as "bacteria": it corresponds to the places where there are bacteria
- white, labelled as "membrane": it corresponds to the places where there could be bacteria but there are none
- orange, labelled as "unsure": it corresponds to places where we can't distinguish between the other classes because the signal is too high or too low

An annotation file corresponding to the aforementioned instructions is attached with the test. It can be opened using VIA (see attached html file):

1. The first question of this test is to convert these annotations to
 - a. masks usable to train a deep learning model
 - b. and to a similar version of these masks that are readable to the human eye.

Please note that the script has to take as input the manual annotations and not use the colors in the dummy image directly as we want here to mimic a real world scenario where the bacteria are not easy to differentiate from the membrane.

We do not recommend using a LLM for this specific question.

2. Then, propose a strategy to train a model to perform semantic segmentation on a dataset composed of thousands of images like this one. The strategy has to include the choice of the model (we want to be able to run the inference in ~1s on a GPU), the choice of the loss, the evaluation protocol and improvement / fallback strategies in case the model is not efficient enough for the most common pitfalls encountered in semantic segmentation.
The more precise your strategy the better 😊

3. In semantic segmentation, one common pitfall is class imbalance. Based on our image, we can see that the pixels labeled as bacteria are underrepresented compared to those labeled as membrane.

Focal Loss was developed to solve this issue:

<https://doi.org/10.48550/arXiv.1708.02002>.

This paper gives a focal loss for binary classification. However, our problem is a multiclass task (we have 3 classes: bacteria, membrane and dust).

- a. Propose an implementation of the Focal Loss for multiclass tasks as a subclass of Pytorch's nn.Module.
- b. Then conduct an experimental analysis (without training a model) showing how your implementation behaves on a toy example, how it changes the importance given to each prediction depending on the different hyper-parameters of the focal loss and how it can be beneficial for our specific use case.

Please in your answer make sure to:

- if using a Python script, include the output png in the archive
- if using a Notebook, have the output cells available